SIEMENS energy

SE GT SU
Kaz Power Plant Upgrade Project Risk Analysis

Job-based hazard analysis is performed in A, B, C, Small projects and, by decision of EHS managers, in all projects.

Job-based hazard analysis and the work permit consist of 5 steps: Steps 1, 2, 3, 4 and 5.

Step 1

EXPLANATION: The existing general potential risks and hazards in the work area are defined.
The relevant risk is ticked. 26 risk groups were defined.

Risk Groups	Relevant Work Permits		Work permits related to the risk groups in Step 3
1- Electric shock or exposure to electric arc		1	ELECTRICAL WORK PERMIT
2- Fall from height		2	WORKING AT HEIGHT PERMIT
3- Falling object from height		3	WORKING AT HEIGHT PERMIT
4- Hot work (oxy-LPG, oxy-acetylene, LPG, torch)		4	HOT WORK PERMIT
5- Welding (electrode, aluminum)		5	HOT WORK PERMIT
6- Ergonomic strain / manual handling		6	JHA
7- Hazardous waste		7	MOTAT Records
8- Use of mobile electrical panel and extension cable		8	ELECTRICAL WORK PERMIT
9- Use of crane, hoist, forklift, compressor, personal carrier, etc.		9	CRANE WORK PERMIT, LIFTING PLAN
10- Use of construction machinery (excavator, truck)		10	EXCAVATION WORK PERMIT
11- Angle grinder use		11	JHA
12- Use of chemical substances (ethyl alcohol, acid, base, epoxy, etc.)		12	JHA
13- Waste generation due to activity (cable, metal, paper, hydrocarbon, etc.)		13	MOTAT Records
14- Explosive areas		14	JHA
15- Exposure to dust, vapor, gas, welding fumes, chemical and biological agents		15	JHA
16- Moving machine parts or work equipment (entrapment, impact, etc.)		16	JHA
17- Slip and trip hazards on the ground		17	JHA
18- Contact with extremely cold or hot surfaces		18	JHA
19- Working near energized open busbars and conductors in low-voltage systems		19	ELECTRICAL WORK PERMIT
20- Working in high-voltage system (≥1000 V)		20	ELECTRICAL WORK PERMIT, EKAD
21- Environmental factors, working in extremely hot & cold climate		21	JHA
22- Hand tool use (hammer, cutting tools, etc.)		22	JHA
23- Use of scaffolding, ladder and platform		23	WORKING AT HEIGHT PERMIT
24- Insufficient lighting and ventilation		24	CONFINED SPACE WORK PERMIT
25- Working in confined or restricted space		25	CONFINED SPACE WORK PERMIT
26- Use of pressure vessel		26	CONFINED SPACE WORK PERMIT

Step 2

Existing work steps should be written one by one and work equipment to be used specifically for the job should be specified on a separate line for construction equipment and subjected to hazard assessment.
Severity and likelihood are entered for each step. The severity/likelihood table uses a 5x5 matrix.
According to the severity and likelihood result, the Risk score should be entered as low, medium and high.
Control measures for each written step must be evaluated, and the risk score should decrease in line with the measures taken/to be taken.

Step 3

Relevant PPE is selected to protect against the risks mentioned in Step 2.
Under each PPE in Step 3, the related risk number from Step 2 is referenced to confirm.

Step 4

This is the approval section of the job hazard analysis.
The site manager/field supervisor and OHS specialist conduct the final review and mark the appropriate section.
The OHS specialist informs the field supervisor/site manager as needed that work permits are required for the relevant work steps.
If 'Not Suitable' is indicated, this hazard assessment must be redone. Work cannot start during this time.

Step 5

The approved job hazard analysis must be communicated to employees. In this section, the person performing or supervising the work conveys the content of Steps 2 and 3 to employees.
Employees acknowledge by writing their name, surname and signature in the spaces provided to confirm they accept the items in Steps 2 and 3.

Work starts after Step 5 is completed.

SIEMENS energy	Job Hazard Assesment (JHA)	Document No:	1	Date: 04.03.2026
		Subcontractor		Personnel Tot.:

Activity : Kaz Power Plant Upgrade Project (Constraction, instalation, Test and commitiong of SS extantion)	Used Vehicles : Manlift, Crane, Truck, Excavator
Location : Basrah Iraq	Used Equipments : Hand Tools, Power Tools, Lifting Equipments, Test Equipments
Work Area : Khur Alzubair Power Station	
Team Member	Uğur Akbulut, Onur Özvatan, Abdurrahman Öğüt, Doğukan Arandı

	POLICE	FIRE STATION	AMBULANCE
NOTE: If after risk evaluation and with all the necessary controls in place; the risk rating (R) is HIGH, the EHS department are to be contacted for guidance on the task/process.			
SIEMENS ENERGY EMERGENCY CONTACT NUMBER +90 533 625 9494	**112**		

STEP 1 / ADIM 1 — **HAZARD / POTANTIAL RISK GUIDE**

HAZARD / POTANTIAL RISK GUIDE

Contact with electric shock or arc electric / Elektrik çarpması veya elektrik arkına maruziyet	x	The use of electrical panel and elektrical cable / Seyyar elektrik panosu ve elektrik uzatması kullanımı	x	Contact with dust, steam, gas,welding smoke, biological & chemical substances / Toz, buhar, gaz, kaynak dumanına,kimyasal ve biyolojik maddelere maruziyet		Working with overtemp & extreme cold ,environmental factors / Çevre faktörleri, aşırı sıcak & soğuk iklim şartlarında çalışma	
Falling from height / Yüksekten düşme	x	The use of cranes, chain hoist, forklift, compressors, personel lift, etc. / Vinç, ceraskal, forklift, kompresör, kişisel taşıyıcı v.b. kullanımı	x	Moving part of the machines or work equipment / Hareketli makina parçaları veya iş ekipmanı (sıkışma, çarpma, v.b.)	x	The use of hand tools / El aletleri kullanımı (çekiç, kesici alet, v.b.)	x
Piece falling from height / Yüksekten parça düşmesi	x	The use of construction equipment (scoop, truck) / İş makinası kullanımı (kepçe, kamyon)	X	Slipping points on the ground / Zeminde kayma takılma noktaları	x	The use of scaffolding, stairs, platform / İskele, merdiven ve platform kullanımı	x
Termal cutting (oxy lpg, oxy acetylene,LPG, torch / Isıl işlem (oksi-lpg, oksi-asetilen, LPG, şaloma)		The use of spiral grinding engine / Spiral taş motoru kullanımı	X	Contact with extremely cold or hot surfaces / Aşırı soğuk veya sıcak yüzeylere temas		Inadequate lighting and ventilation / Yetersiz aydınlatma ve havalandırma	x
Welding (Electrodes,aluminium,) / Kaynak yapılması (Elektrot,Aluminyum)		The use of chemical substances (ethyl alcohol, acid, base, epoxy etc.) / Kimsayal madde kullanımı (etil alkol, asit, baz, epoksi, vb)	x	Near working around low voltage conductives and energized opened bare / Alçak gerilim sisteminde enerjili açık bara ve iletkenlere yakın çalışma	X	Working with closed and circumscribed area / Kapalı ve sınırlandırılmış alanda çalışma	x
Ergonomic strain / lifting heavy / Ergonomik zorlanma / ağır taşıma	x	Produced waste as a result of activity (cable,metal,paper,hydrocarbons.etc) / Faaliyet sonucu atık oluşumu (kablo, metal, kağıt, hidrokarbon , vb)	x	Working with high voltage (≥1000 V) / Yüksek gerilim sisteminde (≥1000 V) çalışma	X	The use of pressure vessel / Basınçlı kap kullanımı	
Hazardous Waste / Tehlikeli Atıklar	X	Explosive Area / Patlayıcı Alanlar		Others / Diğer Çalışmalar		Others / Diğer Çalışmalar	

STEP 2	THIS DOCUMENT IS RECORDED AND STORED BY THE DOING THE JOB

Persons Affected: (e.g.operators,maintenance,visitors,public etc):

Handling: For Siemens Energy internal and subcontractor use only.
Review: If there is a change in working methods and conditions, the form must be renewed.

The Risk Assesment

1. Identify and record for each hazard/risk identified the appropriate rating for severity and likelihood without control.

2. Identify and record the control measures to be used then repeat the assessment with the controls in place.

Severity (S)

1:No Injury
2:Minor Injury
3:Lost time Injury **5:**Death or Permanent Disability

Likelihood of harm (L)

1:Very Unlikely
2:Unlikely
3:Likely
4:Very Likely
5:Certain

Risk Score (R) = S x L

1 to 6 = Low Risk (L) (Acceptable, Monitor & Review)
8 to 12 = Medium Risk (M) (Acceptable With Controls)
>12 = High Risk (H) Unacceptable

Severity \ Likelihood	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15
4	4	8	12	16	20
5	5	10	15	20	25

No	Pathway of activity /	Assessment / Değerlendirme High (H) / Yüksek (Y) Medium (M) / Orta (O) Low (L) / Düşük (D)			Precautions	Assessment / Değerlendirme High (H)		
		Likelihood (L)	Severity (S)	Risk Score(R)		Likeliho od (L)	Severity (S)	Risk Score(R)
1	BEFORE WORK	4	3	12	1-Make sure you have a valid assembly visa before going to the country for work 2-A competent person must be appointed for the specific task. For certain jobs (crane operator, forklift operator, welder, scaffold builder, electrician, mechanical maintenance man, etc.) the necessary vocational training and qualification documents should be requested. Unauthorized personnel should not work outside of the job description. 3-Make sure that you have a health report approved by the doctor for the last year or valid in your country. 4-Customer safety induction training and SE EHS field induction training should be taken before starting to work at the construction site. Learn the emergency management plan and actions from the customer. (Nearest hospital, timing, etc.) 5-Siemens Energy and the customer risk assessment must be properly read and understood before starting work. In addition, the training attendance report must be signed by each employee. 6-Unexpected earthquake, fire, explosion, etc. Keep your passport and some local cash in case of need. 7-The project site has a lot of risks within the scope of the work area, personal protective equipment should be provided for each employee. * High visible clothing * Chin-attached helmets * Head lamp * Work shoes with ankle protection *Ear protection *Safety glasses * Protective work gloves *Protective dust mask (FPP3) * Disposable overalls 8-Daily working hours cannot exceed 11 hours during the day and 7.5 hours at night, excluding rest periods. 9-Seven days of uninterrupted work cannot be done, one day of rest is obligatory within seven days. 10-The annual overtime period cannot exceed 270 hours.	4	1	4
2	GENERAL SITE WORK	4	3	12	1- Fill the "LMRA" form before starting work and get the necessary work permit from the client. 2- When you see the risks that may affect you at the project site, inform the customer and ask for action. 3- Notify the project manager of the situation if the client does not take any action.	4	1	4

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
3	DRIVING - TRANSPORT TO SITE	3	3	9	1-Everyone must have a valid driver's license for the type of vehicle they drive and be medically fit to drive, in accordance with local regulations. 2-Everyone is expected to plan their journeys taking into account distance, road conditions, travel time and weather conditions. 3-Everyone must obey all local traffic regulations and speed limits while driving. 4-No one should drive while under the influence of drugs or alcohol. 5-Everyone should wear a seat belt at all times. 6-No one should use mobile phones or other devices that cause distraction while driving, including in hands-free mode - this is unsafe and unacceptable. 7-Eating, drinking, setting up a radio or GPS device, etc. It is expected to minimize the risk of distraction. 8-No one should drive when tired or suffering from fatigue. 9-Everyone should take adequate breaks to drive at least every two hours. If you feel tired, stop and rest. 10-Anyone who orders and uses external personal transportation services and rents a vehicle must ensure that they meet this Zero Accident Culture behavior and the "Vehicle Safety" Principle. 11-Everyone should ensure that the loads carried are within the vehicle weight and load limits, and that the loads carried are properly secured before departure. 12-While driving, you should be careful about the possibility of wild or domestic animals on the road. 13-The vehicle should be used considering weather conditions such as rain, snow, fog, icing. 14-Tires suitable for seasonal conditions should be used (such as winter tires in cold and rainy weather, summer tires in hot weather)	3	1	3
4	GENERAL SITE WORKS (STOP)	4	3	12	1-Everyone is empowered to apply Stop any time there is a potential risk of harm or non-compliance with Zero Harm Behaviours, Essentials, local Zero Harm programs, or legal requirements. 2-Taking into consideration the severity of the circumstances, Stop! may be issued by anyone as follows: *Immediate direct verbal Stop!, usually involving minor non-compliance, including unsafe acts or conditions that can be easily corrected on the spot without delay. *Immediate notification of Stop! to the responsible person in charge of the work activity or workplace condition for immediate corrective action. 3-Everyone experiencing a Stop! notification from another individual is expected to receive it in a constructive way and as a kind reminder of our Zero Harm Principle that "We Take Care of Each Other." 4-Everyone must respect the Stop!, and it shall remain in force until required corrective measures have been put in place. 5-When in doubt how to resolve a situation leading to a Stop!, everyone must consult their managers and verify the applicability and implementation of required measures.	4	1	4
5	INCIDENT MANAGEMENT	4	3	12	1-Everyone needs to know the Siemens Energy definitions and procedures for reporting incidents and near misses. 2-Everyone should ensure that any incident or near miss they are involved in or witnessed is reported to their manager or supervisor without delay. 3-Anyone who experiences an injury, no matter how minor, or who experiences sudden pain from a work activity should speak up without delay and report the situation to their manager or supervisor. 4-Anyone who has been involved in or witnessed an incident or near miss will support the investigation process and contribute to cause analysis, lessons learned and closure of the incident. 5-Everyone should be familiar with the corrective and preventive measures required following the investigation process and be fully aware of any changes in procedures or risk assessments covering business activities. 6-Everyone should support the implementation of corrective and preventive measures by proactively checking that everyone they work with is fully aware of changes.	4	1	4
6	WORKING UNDER ELEVATED PHYSICAL STRESS AND MANUAL LIFTING/HANDLING	3	3	9	1-Plan for adequate breaks 2-Avoid work that is too frequent or too long, requires extreme distances, or is done with insufficient rest time. 3-Use auxiliary equipment for lifting and transporting. If it is not possible to use the equipment, get help from other employees. 4-Take into account the physiological state of the employee 5-The characteristics of the load such as size, weight, manageability, location of the center of gravity, external and/or internal characteristics shall be taken into account. 6-Features of the study area such as insufficient space, uneven ground and climatic conditions will be taken into account. 7-Appropriate instruction on the correct handling of loads and the hazards arising from misuse 8-Assign only those personnel who are physically fit to perform the task	3	1	3

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
7	WORKING AT HEIGHT	5	3	15	1-Scaffolding installation/dismantling must be done by competent personnel, the necessary documents regarding scaffolding suitability will be submitted by the customer to Siemens Energy before the work. 2-Personnel working on the scaffold must use the necessary PPE (full body protection seat belt, yoyo, lifeline, etc.) 3-Materials and equipment should not be left on scaffolding, access platforms, etc. at the end of the work or during shift change. The risk of objects falling should be eliminated. 4-Work should not be done on scaffolds where guardrails and barriers have not been installed. 5-Only access inspected and approved scaffolding. Scaffolds with green labels are suitable, those with red labels are not suitable. 6-Scaffolding should only be installed or replaced by a specialized company. 7-All scaffolding should be inspected weekly by a trained and qualified scaffolding inspector. 8-Every employee should check the scaffolding to observe obvious deficiencies before going on the scaffolding. 9-Each study should be evaluated on its own and collective protection measures should be given priority. In cases where collective protection measures cannot be taken, precautions should be taken with personal protective equipment and equipment. 10- In the work to be done on the GT, the use of yoyos attached to the lifeline should be a priority. In cases where lifelines need to be dismantled due to the requirements of the lifting operation, the yoyos should be connected to the ceiling crane, anchor points should be used, or the yoyos and the turbine should be anchored at appropriate points to be determined and seat belts should be used.	5	1	5
8	WORKING WITH BUILDING/CEILING CRANE	4	3	12	1-Maintain distance from load 2-Keep transport routes free 3-Use edge protection to prevent damage to lifting gear due tosharp edges 4-Prepare lifting plan if complex crane operations are to beperformed 5-Lifting gear / load rigging shall be visually inspected before use 6-Notify supervisor of defective lifting gear/load rigging 7-Mark the defective lifting gear red and block for further use 8-Always monitor load and crane 9-Use chain hoist for small lifting operations or use chain hoist oncrane hook; use the actual crane for larger operations 10-Do not give any direction commands while the weight isswinging from the crane hook; wait until the weight has cometo a standstill 11-When lifting with the crane hook, take care to ensure that thechain does not become snagged 12-Plan all crane activities in advance and prepare the set downarea 13-Never work under a suspended load. 14-Ensure clear and precise signals between rigger and craneoperator 15-Only the rigger shall give signals to the crane operator 16-Do a walk down of the lift path before crane work and warnpersonnel working in the hazardous area 17-All personnel shall clear the hazardous area if the load is to be moved above work areas 18-Always use the correct device for lifting operations; consultyour supervisor if you are uncertain 19-Never turn your back on the load 20-Use only appropriately trained and qualified personnel 21-Periodic inspection reports should be provided before the overhead crane is used. 22-Before the lifting operations, the visual control of the lifting equipment should be done and periodic control reports should be provided. 23- Only employees with authorized overhead crane operator certificate can use the overhead crane.	4	1	4
9	HAZARD DUE TO ELECTRICAL CURRENT, HIGH VOLTAGE	4	4	16	a-The following precautions will be taken by the customer during Electrical Work Compliance with 5 safety rules: 1. Isolate 2. Lock Out Tag Out 3. Confirm that equipment cannot be re-energized 4. Grounding and short circuit 5. Maintain isolation with adjacent energized elements b-Work permit should be taken. c-Safety instructions must be followed. d-Use only the tools appropriate for the activity. e-Use shielded extra low voltage or safety extra low voltage in confined areas f-Wear appropriate personal protective equipment g-Only electrically trained and certified employees should work in electrical work.	4	1	4

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
10	Working with Forklift and Mobile Cranes	4	3	12	1-Before using forklifts and mobile cranes, periodic control documents, periodic maintenance records, and operator certificates of the employees who will use them should be obtained. 2-Except for the relevant operators, forklifts and mobile cranes should not be operated by other employees. 3-For forklifts, reverse gear warning alarm and strobe warning light will be in working condition. 4-No work should be done in the mobile crane and forklift operation and maneuvering area, and personnel should be removed. 5-Lifting plans should be considered in lifting operations. 6- Forklift transports should be carried out considering the center of gravity of the load. 7-Lifting and conveying personnel should not be done with mobile cranes, overhead cranes and forklifts without equipment complying with the standards. 8-In order to prevent the parts from falling during lifting with forklifts, overhead cranes and mobile cranes, the parts should be lifted with durable and closed systems such as boxes and crates.	2	3	6
11	HOT WORK(e.g., grinding, welding, cutting)	4	3	12	1-Work permit must be taken before work 2-Mark areas and set up barricades if necessary 3-Remove flammable materials from the environment 4-Have an observer in case of fire. 5-Keep fire fighting equipment with you 6-Wear appropriate personal protective equipment 7-Use full-coverage goggles or face visor for cutting, grinding, etc.	4	1	4
12	Field Activities Trips/Falls/Injuries	4	3	12	1- Materials being transported must be stacked in an organized manner. Materials should be stacked in separate areas according to their size and weight, ensuring they do not tip over or collapse. 2- Materials should be transported using appropriate vehicles. 3- Slings used for material transport must undergo visual inspection. Worn slings should not be used. 4- Do not stand under the load while lowering materials with a crane. Do not lift loads exceeding the crane's capacity. 5- Material transport and service vehicles should not be operated by anyone other than authorized personnel. 6- First aid kits and fire extinguishers must be available in the vehicles. 7- Work should not be carried out in rainy, foggy, or heavy weather; weather reports should be checked before work begins."	4	1	4
13	PPE & Equipment	4	3	12	"At the start of work, all personnel's PPE assignment forms are checked using the personnel on-the-job control form. PPE deliveries should be made in the form of notification and receipt. PPE should be checked at least once a month and the checks should be recorded. If employees detect any malfunction or damage to their personal protective equipment, they should inform their supervisor and ensure it is replaced. Minimum PPE; 1- Helmet EN 397 (chin strap for teams working at height) 2- Work shoes EN 345 S3 or S1P (with sole and toe protection), Safety glasses EN 166 (anti-fog), Gloves EN 388 (for employees), EN361 safety harness should be used for work at height. 3- In addition, appropriate personal protective equipment (safety harnesses, welding masks/gloves, dust masks, etc.) according to the nature of the work should be delivered and recorded on the PPE assignment form. 4- Company logo work The outfit should be worn."	4	1	4

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
14	Installation of Switchyard Equipment	4	3	12	"Before work begins, safety jump distances must be evaluated. The power should be cut off at points within the safety distance, and the 5 safety rules should be applied. 1- Devices should be lifted using a mobile crane that has undergone periodic checks and has the appropriate and inspected legal documents. 2- There must be a slinger connecting the devices and a signalman directing the operation. 3- The crane legs must be fully extended to the solid ground. In case the legs sink, materials that will expand the surface area should be placed under the legs. 4- Material guiding ropes should be used. 5- A crane work permit must be filled out. 6- No personnel should enter under the material under any circumstances. The work area should be restricted and warning signs should be placed. 7- The operation should be carried out taking weather conditions into consideration. 8- For all work to be done with the crane, the ""CRANE WORK CHECKLIST"" must be filled out and approved. 9- During the lifting operation, the area should be restricted, and the area should be fenced off to prevent entry and exit. No one other than the signalman/slinger should be in the area. The slinger/signaler should under no circumstances go under the lifted load." 10- During assembly, it should be checked whether simultaneous (top-bottom) work is being carried out. Any simultaneous work should be stopped immediately. 11- Slingers guiding during assembly should not be under the pylon/beam/load. 12- Climbing the pylon should be done using a vertical lifeline and/or yo-yo, and walking on the equipment should be done using a horizontal lifeline.	4	1	4
15	WORKING IN CONFINED SPACES(inadequate evacuation and rescue possibilities, poor communicationto the outside, risk of panic attacks, poor ventilation conditions,insufficient room to move)	4	3	12	1-Before starting work, the necessary work permit must be obtained. 2-Perform a visual inspection of tagging and locking for correct application before entering confined spaces 3-Use only safe extra low voltage (SELV) or shielded extra low voltage (PELV) electrical equipment. 4-Assign a watchman on the entrance manhole and maintain constant contact with the watchman. 5-Check the atmosphere (oxygen concentration) before starting work, keep rescue equipment ready if necessary 6-Maintain order and cleanliness in confined spaces during maintenance work 7-It is forbidden to work alone; Must have at least 2 employees 8-Appropriate persons should be assigned to the job. 9-Respirator should be used if necessary 10-Ambient air measurement should be performed before entering the enclosed space. 11-When accessing the pier, only the access point provided for this purpose should be used. 12-Access and exit points should be clearly defined. 13-Emergency exits must be kept open at all times.	4	1	4
16	Portable ladder usage	4	3	12	1- Ladders must be selected according to the height at which the work will be performed. 2- Ladders must have a CE Certificate and must have manufacturer information and a label indicating CE standards, and must comply with the EN131 standard. 3- Work should not exceed 30 minutes on ladders, and heavy work requiring prolonged use of equipment with both hands (cutting with a stone cutter, etc.) should not be performed. 4- Ladders should be checked before each use. Inspection cards should be available and checked at least once a week. 5- On type A ladders, the securing strap or locking mechanism must be completely locked when the ladder is opened. 6- Ladders with deformed, broken, or bent legs should not be used for work. 7- Contact should be made at least at 3 points on the ladder during work. 8- Work should not be done on the top two steps of the ladder. 9- The ladder to be used for climbing must be sufficiently strong and secured by securing it at a suitable point. In cases where securing is not reasonable or possible, ladder securing should be done according to the manufacturer's instructions." "These tools should be used."	4	1	4

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
17	WORKING WITH HAND TOOLS/POWER TOOLS /MEASURING EQUIPMENTS	4	3	12	1-Only use tools which are appropriate or intended for the activity 2-Visual inspection before use 3-Safety instructions in the instruction manual shall be observed 4-Wear appropriate personal protective equipment 5-Use low-vibration equipment or schedule sufficient breaks 6-Secure tools in such a way that slipping away during operationis prevented 7-Shield adjacent work areas if injuries to others are possibledue to flying particles or slung parts check inspection tag 8-Safety instructions in the instruction manual shall be observed 9-Only use tools which are appropriate or intended for the activity 10-Check to ensure that safety equipment is intact 11-Use only appropriately trained and qualified personnel 12-Lines (electrical, pneumatic, compressed and process air) shallbe safely routed and secured 13-If hydraulic fluid is used, follow the instructions in the safety data sheet or the operating instructions 14-Wear appropriate personal protective equipment 15-Use whipcheck when using hydraulic and pneumatic systems.	4	1	4
18	HAZARD DUE TO FALLING	4	3	12	1-Installation of scaffolding or barriers, only access scaffoldingthat has been inspected and approved 2-Observe work instruction on use of PPE to prevent falling! 3-Use PPE to prevent falling	4	1	4
19	HAZARD DUE TO SLIPPING, STUMBLING AND FALLING ON SAME LEVEL	3	3	9	1-Lay cables outside the hazard zone 2-Materials and tools which are not required shall be removed from the work area 3-Traffic routes shall be kept clear 4-Keep the work area clean and tidy 5-Make sure that there is no deformation, lubrication or gap on the work area floor, and take the necessary precautions, if any.	3	1	3

No	Pathway of activity /	Assessment / Değerlendirme High (H) / Yüksek (Y) Medium (M) / Orta (O) Low (L) / Düşük (D)			Precautions	Assessment / Değerlendirme High (H)		
		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
20	HAZARD DUE TO WORK WITH BUILDING/CEILINGCRANE/LIFTING GEAR/LIFTING ACCESSORIES AND WORK WITH SUSPENDED LOADS	4	3	12	1-Barricade off work area if necessary 2-Use chain hoist for rolling parts in and out 3-Use crane for larger operations, also use chain hoist ifnecessary 4-Use provided devices 5-Use edge protection to prevent damage to lifting gear due tosharp edges 6-Use all load attachment rigging correctly 7-Do not work under suspended loads 8-Lifting gear / load rigging shall be visually inspected before use 9-Check inspection tags for validity 10-Mark defective lifting gear and block for further use 11-Notify supervisor of defective lifting gear/ slings 12-Maintain distance from load 13-Keep transport routes free 14-Do a walkdown of the lift path before crane work and warn personnel working in the hazardous area 15-Plan all crane activities in advance and prepare the setdownarea 16-Never turn your back on the load 17-Ensure clear and precise signals between rigger and crane operator 18-All personnel shall clear the hazardous area if the load is to belifted above work areas 19-Always use the correct device for lifting operations; consultyour supervisor if you are uncertain 20-Keep personnel who are not required out of the hazardousareaUse only appropriately trained and qualified personnel 21-Wear appropriate personal protective equipment (yellow safetyvest if necessary)	4	1	4
21	HAZARD DUE TO ENTERING, EXITING Confined Space	4	3	12	1-When accessing scaffolding, use only the access pointprovided for this purpose 2-Clearly define access and exit points 3-Keep emergency exits clear at all times 4-Keep work area free from oil	4	1	4
22	HAZARD DUE TO UNRESTRICTED AREA	3	3	9	1-Barricade off work area if possible. 2-Keep personnel who are not required out of the hazardous area 3-Use only appropriately trained and qualified personnel	3	1	3
23	Working with Manlifts	4	3	12	1- Manlifts should only be used by experienced and authorized/trained operators. A trained personnel familiar with the ground rescue plan must be present while working at height on the manlift. 2- Visual checks must be performed by the user before each operation. 3- Proper use of PPE (Personal Protective Equipment) in accordance with working at height regulations must be ensured. 4- Before starting work, the ground must be checked to ensure it is stable and level. Work should absolutely not be done on slippery surfaces. When changing positions, there should be no obstacles, gaps, cables, or hoses along the route that could cause tipping; the scissors must be in the closed position. 5- Wheel locks and support legs must be used to prevent the manlift from moving during operation. 6- The capacity of the basket (Safe Working Load) must not be exceeded; it should only be used to lift workers and hand tools. Maximum load information must be clearly visible on the equipment. 7- Limit switches must not be modified or deactivated. 8- Do not climb on or stand on the guardrails of the manlift. 9- At the end of the shift, the machines must be..." The machines must be locked. The machines must be placed on a solid and level surface where there are no obstacles or movement. The platform must be lowered, the ignition key removed and kept in a place inaccessible to unauthorized persons, and chocks must be placed under the wheels. 10- If welding is to be performed during operation, the welding frame must not come into contact with the machine. External grounding of the body must be provided. 11- When working on the manlift, attention must be paid to exposed power cables and electrical trays, and a safe working distance must be maintained.	4	1	4
24	HAZARD DUE TO INSUFFICIENT KNOWLEDGE IN FIRE EVACUATION	4	3	12	Necessary instructions will be communicated to the staff in accordance with the Customer Emergency procedure.	4	1	4

No	Pathway of activity /	Assessment / Değerlendirme High (H) / Yüksek (Y) Medium (M) / Orta (O) Low (L) / Düşük (D)			Precautions	Assessment / Değerlendirme High (H)		
		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
25	HAZARD DUE TO FIRE OR EXPLOSION	4	3	12	1-Only use tools which are suitable or intended for the activity 2-Remove hoses from combustible gases and oxygen ordisconnect from supply point when work is interrupted 3-Maintain a distance of at least 3 meters between fuel gasesand oxidation gases or provide for a fire-resistant partition 4-Safety instructions in the instruction manual shall be observed 5-Visual inspection before use 6-Remove fire loads, assign fire watches, provide forextinguishing agents 7-ATTENTION! Liquid gas cylinders (propane or butane) shall bestored at least 3 meters from ALL other types of gas cylinders 8-NO SMOKING in oil storage area 9-Wear appropriate personal protective equipment	4	1	4
26	HAZARD DUE TO FALLING LOADS	4	4	16	1-Barricade off work area 2-Suspend load in accordance with drawing specifications 3-Keep personnel who are not required out of the hazardous area 4-When working at height, all hand tools should be kept in the keeper.	4	1	4
27	Waste Management	3	3	9	1-Waste must be sorted according to waste group (metal, cable, wood, paper, etc.). 2-Waste must be sorted according to waste class and disposed of in the waste area.	3	1	3

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
28	HAZARD DUE TO POOR ILLUMINATION	3	3	9	1- Illumination must be increased according to conditions 2-Lighting that is appropriate to the situation shall be provided	3	1	3
29	Lifting Operation	4	3	12	1- The weight of the material to be lifted must be known before lifting, and the crane must be selected according to the weight of the material to be lifted. 2- The lifting accessories to be used must have appropriate capacities and must be in a safe condition without deformation. 3- Unauthorized personnel should not be present in the work area. 4- The crane must undergo periodic inspection, the operator must have a certificate and it must be checked. 5- The lifting area must be marked and a warning sign must be placed. 6- The crane's equipment must be checked and not overloaded beyond its tonnage. 7- The crane's legs must be extended, the ground checked and ensured to be level, and the crane's outriggers and pads must be used to adapt to the ground. 8- Weather risks must be assessed (Reminder: In case of winds exceeding 50 km/h, all operations are suspended). 9- The necessary distance between the load and the operators must be provided and the personnel must be informed. 10- The operation of the signal warning lights must be checked. 11- Absolutely no one should go underneath the material. If necessary, a rope/cord should be used for guidance. 12. When securing the material, if the material has sharp edges, cardboard or similar material should be placed under the material to prevent damage to the slings; care should be taken. 13. No load exceeding the tonnage of the slings should be attached. 14. The lifting operation must be controlled by the signalman. If visual contact is not possible, communication between the crane operator and the signalman should be ensured by providing them with a radio. 15. All personnel performing the operation must wear all necessary appropriate personal protective equipment. 16. When bringing the panels into the control building, they must be lowered due to the door height; this operation must be done with the help of a crane. 17. The slings used to lift the panels must be suitable for the weight of the panels, and deformed slings must absolutely not be used.	4	1	4
30	Cable pulling	4	3	12	When removing the cable reel from the truck, a sling appropriate for the weight of the material must be used. A crane operator must be used, the crane's lifting apparatus must be checked, the slinger-signaler must direct the operation, the crane's outriggers must be sufficiently spread apart, and pads must be placed under the outriggers to ensure stability on the ground. No one should be in the area of operation, no one should enter under the material, a guide rope must be used, the necessary warnings must be given, and a lookout must be present. The cable reel must be set up on a level surface using a cable stand suitable for the tonnage of the cable reel. There must be a sufficient number of personnel at the head of the cable reel to turn the reel. No one should stand in front of the cable reel when the fiber guide is being pulled through the laid pipes, as the end is hard. The cable pulling device must be anchored to the ground with dowels. The area must be enclosed with railings and barriers. All ladders, platforms, or scaffolding used for cable pulling must be inspected by a competent person, certified with a green card, and comply with standards. The cable pulling device and reel transport stands must have periodic inspection reports issued by competent institutions within the last year.	4	1	4
31	HAZARD DUE TO UNINTENTIONALACTIVATION/INCORRECT OPERATION OFSYSTEMS/COMPONENTS	4	3	12	1-Permit to work has been issued 2-NOTE: Permit to work shall be prepared and issued only byperson tested and authorized by the customer 3-Ensure communication with all participating persons beforeactivation and operation 4-Deactivate pumps before checking oil strainers.	4	1	4
32	HAZARD DUE TO HANDLING OF INSULATING MATERIAL	4	3	12	1-Keep personnel who are not required out of the work area 2-Wear protective clothing and respirator masks. 3-Use fall protection equipment when necessary.	4	1	4
33	HAZARD DUE TO UNAUTHORIZED PERSONS/ OTHER SUBCONTRACTORS IN WORK AREA	4	3	12	1-Mark areas and barricade off if necessary or set up warningsigns. 2-Inform subcontractors of risks	4	1	4

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likeliho od (L)	Severity (S)	Risk Score(R)
34	Assembly of Primary Equipment - Crane operations, use of slings and shackles.	4	3	12	1- Lifting accessories must be stored in a way that prevents damage or deterioration. 2- Ropes, slings, and chains must be replaced when deformation occurs. Their certifications must be recorded. 3- Slings and shackles capable of supporting 1.25 times the weight of the load to be lifted must be used, and the slings and shackles must have label certificates. Worn or damaged equipment without labels should not be used. 4- Slings must be attached at the appropriate angle. 5- Protective pads must be used at points where they come into contact with sharp surfaces. 6- Equipment must be checked for deformation before each operation. 7- Personnel must use double lanyards when climbing pylons to install beams and remove slings. It must be absolutely ensured beforehand that the areas where the lanyards will be attached are securely created. 8- Long slings should be used to ensure that the slings are close to the pylon during beam installation, preventing personnel from walking on the beam. If short slings are used, personnel will attach to the yo-yo, which is fixed to the crane hook, to remove the slings. 9- Ascending the pylon should be done using a vertical lifeline and/or yo-yo / Walking on the equipment should be done using a horizontal lifeline.	4	1	4
35	Current Transformer, Voltage Transformer, Surge Arrester Installation of Supports	4	3		1- Before starting ground assembly work, the work area must be prepared and organized; factors that could cause slipping, tripping, and falling must be eliminated. 2- Sling attachment at the supports should be done at the top, and a protective pad or similar piece should be used under the sling to prevent damage. 3- Personnel must work with a safety harness. 4- After the support assembly is complete, the sling should be retrieved from the top using an appropriate type of ladder.	4	1	4
36	WILD ANIMAL BITES / EXPOSURE TO DISEASE-CARRYING VOLATILES	2	4	8	In mild injuries, the wound should be washed with soap and cold water for 5 minutes. The wound should be covered with a clean cloth. If there is serious injury and bleeding, the bleeding should be stopped by applying pressure to the wound with a clean cloth and support should be received immediately from 112. Panzeir status for wild animals belonging to the study area should be learned from health centers. The worker should be transferred to the hospital in possible snake bites and it should be ensured that snake serum is intervened against snake bites. • Comprehensive information on common local carriers and diseases should be collected. If necessary, special expertise should be acquired at the first stage. • Personal behavioral training should be provided. (e.g. food disposal, clothing, fly repellents) • Mechanical vector control measures should be taken. (eg mosquito screens, mosquito nets) • Reproduction control should be done. (eg disposal of still water) • Chemical or biological vector should be controlled. (applied by experts) • If there is a deficiency in routine vaccination, it should be completed, if possible and necessary, additional immunization should be done. • While driving, you should be careful about the possibility of wild or domestic animals on the road.	1	4	8

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)
37	Cable laying	4	3	12	"When removing the cable reel from the truck, a sling suitable for the tonnage of the material must be used. The crane must be operated by an operator, the lifting apparatus of the crane must be checked, the sling operator/signaler must direct the operation, the crane's outriggers must have sufficient clearance and pads must be placed under the legs for adaptation to the ground. No one should be in the operation area, absolutely no one should go under the material, and a guide rope must be used, necessary warnings must be given, and a lookout must be present. The cable reel must be set up on a flat surface using a cable stand suitable for its tonnage. There must be enough personnel at the head of the cable reel to turn the reel. No one should stand in front of the fiber guide when it is being driven through the laid pipes because its tip is hard. The cable pulling device must be fixed to the ground with dowels. The area must be enclosed with railings and barriers. The cable pulling device and reel carrying stands must have a periodic inspection report issued by competent institutions within the last year. Arc protection must be worn and insulating mats must be laid in the channel where energized MV cables are located." "Cable reels should be fixed inside the manholes, and the cable should be pulled through them. Manholes should be cordoned off with tape and covered at the end of the shift. The number of personnel for cable pulling should be appropriate for the operation, and communication should be via telephone. Cable cutters should be used for cable cutting. Personal protective equipment should be worn and used. (Helmet, work gloves, safety glasses, reflective vest, work shoes) Boots should be worn due to the accumulation of rainwater and sewage water inside the manhole."	4	1	4
38	Switchyard Circuit Breaker Installation Work	4	3	12	Before work begins, site officials must conduct a site survey to identify hazards and risks. Work should not commence without taking necessary precautions. -Before mechanical assembly work begins, appropriate work permits must be obtained and work permit requirements must be met. -Work at height requirements specified in the "Loading the primary material to the site" work step must be ensured. -Crane use and safety precautions specified in the "Mechanical assembly work" work step must be ensured. -Materials will be lifted using 2 slings and by lowering them from the upper section; precautions will be taken against material tipping, tilting, etc. The power must be cut off at points within the safety distance, and the 5 safety rules must be applied."	4	1	4
39	Landscaping and Excavation Works	4	3	12	1- The licenses, maintenance records, and other legal documents for trucks and construction equipment (JCS, Loader, Excavator, etc.) to be used on site must be provided. 2- Drivers' personal documents must be submitted, and they must receive on-the-job training. 3- An additional observer must be present in the maneuvering and working area of trucks and construction equipment. 4- Workers must be prevented from entering the truck's blind spot, and training must be provided to workers on this subject. 5- No one should be near the truck when loading excavated material onto it. 6- Excavated material must be transported to the area indicated by the customer. 7- Equipment must be available on trucks to prevent excavated material from spilling.	4	1	4
40	General field activities - Working near energized points	4	3	12	1- Energized areas must be restricted using appropriate barriers, tape, etc. 2- Work must be performed using equipment that will not approach the safety distance (excavators, etc.). The safety distance must be calculated prior to work. 3- Energy shutdown must be requested for work within the safety distance, and the 5 safety rules must be applied. 4- Employees must be trained on energy points. 5- Employees must not raise materials they are holding upwards	4	1	4
41	Emergency situation	4	3	12	1-An emergency action plan should be organized and communicated to all employees. 2-Teams that will respond to emergencies should be identified (first aid team, fire team, search and rescue team, evacuation team, etc.) and trained, and the training provided should be recorded. 3-The list of names and contact information for emergency response teams must be posted in a visible location in the work area and mobilization area. 4-The contents of the first aid and emergency kits must be determined by the occupational safety specialist and technical authorities, procured, and kept on site. 5- Every worker on the construction site should be given an ID card. The back of the ID card should contain the project emergency numbers, and the front should contain the employee's identity information and the training they have received.	4	1	4
42	Commissioning	3	4	12	1-Commissioning must be made by appropriate qualification person. 2-Working areas shall be marked and barricaded off. Forbidden for other person. 3-Take work permit from customer, before start the commissioning.	2	3	6

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likeliho od (L)	Severity (S)	Risk Score(R)
43	Working with Chemicals	4	3	12	1-Chemicals that are harmless to health should be chosen. 2-It should be used taking into account the MGBF. 3-Chemical wastes should be disposed of according to legal and local rules. 4-Spilled chemicals should be cleaned immediately with absorbent and cleaning materials. 5-If hydraulic fluid is used, it should be used in accordance with the MSDS form. 6-It should be stored at a minimum level. 7-In order not to be exposed to chemical fumes, dust or fog, suction of the air in the environment, cleaning the air or protective masks should be used. 8-Personal Protective Equipment specified in the MSDS of chemicals should be used. 9-If chemicals spill or leak into an environment, they should be cleaned with absorbent kits.	2	3	6

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		Likelihood (L)	Severity (S)	Risk Score(R)		Likelihood (L)	Severity (S)	Risk Score(R)

STEP 3 /	PPE & Equipment															
PPE & Equipment	LOTO	EN 397 Helmets	EN 352 Ear Plug	EN 345 S3 Safety Shoes	EN 388 Mechanical	EN 407 Heat-Tolerant Glove	Protective Clothing	EN 166 Eye wear	EN 166 Face Shield	EN 407 Thermo Safety Glove	EN 361 Full Body Harness	Electric Arc Protective Clothing	Electrical Safety Gloves	EN 136 Dust Mask	Reflective Vest	Fire Extinguisher	Other
Path the Follow No	Electrical work if neccessary	All	All	All	All		All	All			Working at height	Electrical work if neccessary	Electrical work if neccessary		All	All	

STEP 4 /	HEAD of PROJECT MANAGER	QUALITY MANAGER	PROJECT MANAGER		EHS	
Approval / Onay	Name Surname: UĞUR AKBULUT Date: Signature Phone	Name Surname: ABDURRAHMAN ÖĞÜT Date : Signature Ogut Abdurrahman Digitally signed by Ogut Abdurrahman Date: 2026.03.04 11:53:38 +03'00' Phone	Name Surname: ONUR ÖZVATAN Date: Signature Phone	SUITABLE UNSUTABILE	Name Surname : DOĞUKAN ARANDI Date: Signature Arandi Dogukan Dijital olarak imzalayan Arandi Dogukan DN: cn=Arandi Dogukan, o=Siemens, email=dogukan.arandi@siemens-energy.com Tarih 2026.03.04 11:18:39 +03'00' Phone	SUITABLE UNSUTABILE

STEP 5	TRAINER/SIGN	TRAINING AREA	TRAINING DATE
	Name Surname	Signature	

Personnel Signatures				
		6.	11.	16.
		7.	12.	17.
		8.	13.	18.
	4.	9.	14.	19.
	5.	10.	15.	20.

I guarantee that i will follow the rules, mentioned on the precautions part of risk assesment form, and also under negative conditions i will stop working in a safe way,then inform my project manager